PROJECT TITLE:

Coexistence of native-exotic communities in disturbed dendritic networks

MAIN AIM:

Inferring the composition of species assemblages of a fish metacommunity accounting for abiotic, biotic, migration and fragmentation factors.

MAIN TASKS:

1) DATA (Ramon and Lucia): P/A and density species matrix data, pairwise distance-altitude matrix data for Guadalquivir basin, and abiotic-migration-biotic factors raw data

2) MODELS (Alex and Carlos): Infer distributions of presence accounting abiotic, biotic, migration, and fragmentation factors

3) MODEL-DATA COMPARISON: Approximate Bayesian computation (ABC) or likelihood methods to infer the factors that best explain the data.

4) DRAWING the Guadalquivir DENDRITIC NETWORK: Draw dendritic network using real world coordinates together with the reservoirs. Put it as a figure 1 to illustrate clearly the approach.

5) BIBLIOGRAPHY. We have already a folder with references. All references will be included in this folder and in the $.bib$ file.

6) REPRODUCIBLE RESEARCH DOCUMENT: Jupyter notebook to make all the steps easily reproducible.

Coexistence of native-exotic communities in fragmented dendritic networks.

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Keywords: α, β and γ-richness,
fragmented landscapes, environmental and interaction selection,
migration, metacommunity dynamics,
Type of Article: Article
Number of figures: X; Number of tables: YY
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Abstract

2 .

1 Introduction

This Intro is taken from the "diversification in dendritic networks" draft: Which is the relative importance of abiotic, biotic, migration and fragmentation to predict the presence of species assemblages? Why do native species coexist? Do they go to extinction once the exotic are introduced?

Freshwater communities of fish in lakes are among the richest vertebrate species assemblages on Earth (Lévêque *et al.*, 2007; Vadeboncoeur *et al.*, 2011), but they are also among those with the steepest rates of species loss (Vonlanthen *et al.*, 2012; Seehausen *et al.*, 1997) (other refs?) due to eutrophication, climate change, invasive species and the interaction of these factors (Vörösmarty *et al.*, 2010). Yet, surprisingly little theory and even less data is available to predict how the effects of these environmental factors depend on the history and mechanism of assembly of freshwater communities. Lake fish communities range from little isolated entirely immigration-assembled communities to nearly entirely speciation-assembled, with all intermediate combinations of both processes (Wagner *et al.*, 2014). There are many reasons to expect that communities assembled by different mechanisms can respond very differently to environmental change.

Freshwater bodies in general, and lakes in particular, support exceptionally high numbers of endemic species per unit area. Several examples have shown that diversity in these systems can suddenly collapse when ecosystems are perturbed, and it does often not recover after ecosystem restoration (refs for Lake Victoria; Swiss lakes; Laurentian Great lakes; Lake of Mindanao). Very little is known about how environmental change factors interact with each other and with the assembly history of species communities to determine loss of diversity and consequences for ecosystems. Freshwater ecosystems, and large lakes in particular, are strongly geographically isolated for organisms that lack terrestrial or airborne dispersal phases such as fish. Large lakes have unique and diverse habitats that provide resources for large and very diverse populations, but colonization of lake-specific habitats like the pelagic and profundal zones often requires in-situ evolution and speciation because river fish can rarely colonize such habitats and lakes are often extremely strongly isolated from other lakes. Lakes hence function like isolated oceanic islands where in-situ evolution is a major source of biodiversity (Hudson *et al.*, 2011; Wagner *et al.*, 2014).

Theory suggests that the assembly of communities by in-situ evolution and by dispersal from a regional pool make very different predictions for richness and abundance. For instance, whereas the species-area curve in dispersal-assemblies typically has a slope around 0.3, it is much steeper (near 1) for communities assembled by speciation (Rosenzweig, 2001; Wagner *et al.*, 2014). Theory and data suggest that locally evolved species are often more abundant than others (Rosindell & Phillimore, 2011). Explain the concept of a dendritic network (Campbell Grant *et al.*, 2007). Yet, little is known about how the topology of 3D networks (i.e. dendritic networks in a high-relief landscape) change the assembly process of endemic and non-endemic species in metacommunities. While clearly a question of considerable relevance to conservation in general, this is especially pressing for freshwater ecosystems. Their heavy reliance on evolution for the building of species diversity is expected to influence the ways these ecosystems respond to environmental change. One might for instance expect more dramatic responses in speciation-assembled communities because of the steeper scaling of richness and abundance with area and the often much larger per area-richness (Wagner *et al.*, 2014). Another important difference is that in speciation-assembled communities even transient and reversible perturbation of niche structure can lead to irreversible collapses of species diversity within just a few generations (Seehausen *et al.*, 1997; Vonlanthen *et al.*, 2012).

Here we develop a niche modeling to predict species assemblages of fish communities in 3D dendritic networks accounting for abiotic, biotic, migration and fragmentation factors. We focus on fish but our model may be applicable more widely to lacustrine biota. However, fish account for more than 60% of all endemic animal species in lakes globally (Vadeboncoeur *et al.*, 2011). At the same time fish represent critical links in aquatic food webs. To parametrize our model and test its utility, we make it spatially explicit and apply it to the network of large subalpine lakes of central Europe, a hotspot of freshwater fish endemism in Europe (Kottelat & Freyhof, 2007). Our empirical data includes >20 large lakes and connected rivers located in three archipelagos of large subalpine lakes north (Rhine), west (Rhone) and south (Po) of the central Alps. In each archipelago we have sampled a similar range of lakes in terms of size, elevation but the archipelagos have distinct assembly histories. The communities of all lakes assembled within the Holocene, i.e. in the past 15000 years. But whereas on the South-side of the central Alps, there was rapid recolonization by many lineages from nearby southern refugia with only few glacial relict species surviving, northern and western lakes were probably colonized rapidly by just a handful of cold water fish lineages that may have survived the Pleistocene in near refugia, but then were colonized by all the other lineages only slowly from distant eastern (Black Sea) refugia. They retained a diverse set of cold-water adapted glacial relicts, many of which diversified into ecologically diverse lake endemics. As a consequence, lake fish communities range from entirely immigration assembled to those with large contributions of speciation. Locally evolved species fill ecological key roles in most northern and some western lakes, but not in the southern lakes. We compare data with model predictions of species richness, endemism and radiation.

some references to add for an introduction on 2D river networks: (Honnay *et al.*, 2001; **?**; **?**; **?**)

To generate theoretical expectations for diversity in each lake, we developed an individual-based stochastic simulation model building on recent evolutionary extensions of island biogeography theory (Rosindell & Phillimore, 2011, hereafter $R\&P$). We are asking how richness scales with isolation and lake size, and how the assembly process affects this scaling and also the abundances of species. Island biogeography theory has been almost exclusively tested in terrestrial island systems. This is the first evolutionary island biogeography model for freshwater systems. We wish it will help understand the evolutionary and ecological dynamics of lacustrine species assembly including the formation of endemism and radiations. We expect that our results will also contribute to predicting and explaining variation in the susceptibility of ecosystems to environmental change.

MAIN RESULTS Our modeling framework can be extended in several directions in the future, including predicting intraspecific genetic and trait diversity and the genetic distinctiveness of populations. With the increasingly wide access of Next Generation Sequence methods for non-model organisms, population genomic data can now easily be collected to test the predictions. Other possible extensions are to merge our island biogeography model with whole-island regression and habitat-unit models in 3D (i.e., lake depth or an ecological gradient) to study the effects of habitat diversity and lake size separately, with habitat occupation models (Guisan & Rahbek, 2011) and with trophic theory of island biogeography (Gravel *et al.*, 2011).

2 Methods

Brief description of the different sections

2.1 Data

Brief description of the sampling effort and mention figure 1, the dendritic network from the Guadalquivir basin

2.2 Model

We consider a dendritic landscape consisting of N sampled lakes...

2.3 3D dendritic networks

We generate landscapes consisting of sampled sites with a spatial location given by x_i, y_i and z_i, with z_i representing the altitude (Figure 1 SHOW A 3D DENDRITIC NETWORK). The watercourse distance between each pair of sampled sites was calculated as... The third dimension, z_i represents the elevation values from the lakes...

2.4 Estimating the probability of presence

Explain the function used to account for abiotic, biotic, migration and fragmentation...Classification, Probability of presence function and the different scenarios.

2.5 Abiotic factors

2.6 Biotic factors

2.7 Migration

We frame our hypothesis in dendritic networks using elevation and the hydrological distance between each pair of nodes (i.e., lake-stream, stream-stream, stream-lake). We start by describing dispersal dynamics in the simplest scenario with a 2D network. Dispersal rate between two nodes is a function of their hydrological distance. This leads to the dispersal rate of species k from lake j to the lake receiving the migrant, the lake i

$$m_{ij}^{k} = m \left(\frac{1}{d_{n,j} + \sum\limits_{n=1}^{\mathcal{N}-1} d_{n+1,n} + d_{i,n+1}} \right), \tag{1}$$

where $d_{n,j}$, $d_{n+1,n}$, and $d_{i,n+1}$ are the hydrological distances between lake j and node n, node n and node $n+1$, and node $n+1$ and lake i, respectively, and m is the intensity of dispersal rate. An extension of equation (1) to a 3D network considers dispersal rate not only as a function of the distance, but also as a function of elevation distance between lakes or nodes in the network. This leads to the dispersal rate of species k from lake j with elevation e_j to lake i with elevation e_i

$$m_{ij}^{k} = m \left(\frac{1}{d_{n,j}^{x} + \sum\limits_{n=1}^{\mathcal{N}-1} d_{n+1,n}^{x} + d_{i,n+1}^{x}} \right), \tag{2}$$

with x

$$\begin{cases} x = 1, & e_{em} \geq e_{im}, \\ x = \frac{(1+c(e_{im}-e_{em}))}{1}, & e_{im} > e_{em}, \end{cases} \tag{3}$$

where c is the upstream migration cost between two nodes, e_{em} and e_{im} represent the elevation of the emigration (i.e., e_j, e_n, e_{n+1}) and immigration (i.e., e_i, e_n, e_{n+1}) nodes. We note that equation (2) is the same than equation (1) when c = 0, that is, when upstream migration cost is not present as part of the dynamics (i.e., from 3D to a 2D network). We consider downstream migration has no cost and use the same watercourse distance between two nodes (i.e., x = 1). At a given upstream migration cost, the larger the difference in elevation between two nodes the lower the dispersal rate between these two nodes (Figure 2).

2.7.1 Patterns in the data: distributions of presence, Over-under presence species rank, α-, β- and γ-richness, isolation-richness plots...

2.7.2 α-, β- and γ-richness

2.7.3 Isolation AND SPECIES-AREA(VOLUME) CURVE

We calculate isolation of lake i as the sum of all distances between the focal lake i and all other lakes in the network. This leads to the isolation of lake i,

$$I_i = \sum_{j=1}^{S} \left(d_{n,j}^{x} + \sum_{n=1}^{\mathcal{N}-1} d_{n+1,n}^{x} + d_{i,n+1}^{x} \right), \tag{4}$$

where $d_{n,j}$, $d_{n+1,n}$, and $d_{i,n+1}$ are the hydrological distances between lake j and node n, node n and node $n+1$, and node $n+1$ and lake i, respectively, and S is the number of lakes in the network.

Plot VOLUME x SPECIES curve for the Updated data. Then we will be able to calculate the parameters that best fit to the empirical pattern, like the $\mathcal{K}$ parameter for cladogenesis

2.8 Simulations

Briefly discuss the protocol followed – mentionthe jupyter notebook as a reproducible research document : Simulations were done with all sampled species across all sampled sites. Subsequently we ran simulations following equation (2)... Results plotted in xx and xx were obtained after running X replicates per cost, c,...

3 Results

4 Discussion

5 Results

References

1.

Campbell Grant, E. H., Lowe, W. H. & Fagan, W. F. (2007). Living in the branches: population dynamics and ecological processes in dendritic networks. *Ecology Letters*, 10, 165–175.

2.

Gravel, D., Massol, F., Canard, E., Mouillot, D. & Mouquet, N. (2011). Trophic theory of island biogeography. *Ecology Letters*, 14, 1010–1016.

3.

Guisan, A. & Rahbek, C. (2011). SESAM – a new framework integrating macroecological and species distribution models for predicting spatio-temporal patterns of species assemblages. *Journal of Biogeography*, 38, 1433–1444.

4.

Honnay, O., Verhaeghe, W. & Hermy, M. (2001). Plant community assembly along dendritic networks of small forest streams. *Ecology*, 82, 1691–1702.

5.

Hudson, A. G., Vonlanthen, P. & Seehausen, O. (2011). Rapid parallel adaptive radiations from a single hybridogenic ancestral population. *Proceedings of the Royal Society of London B: Biological Sciences*, 278, 58–66.

6.

Kottelat, M. & Freyhof, J. (2007). *Handbook of European freshwater* *fishes*. Publications Kottelat, Cornol, CH. ISBN 978-2-8399-0298-4. URL http://agris.fao.org/agris-search/search.do?recordID=US201300126031.

7.

Lévêque, C., Oberdorff, T., Paugy, D., Stiassny, M. L. J. & Tedesco, P. A. (2007). Global diversity of fish (Pisces) in freshwater. In: *Freshwater Animal Diversity Assessment* (eds. Balian, E. V., Lévêque, C., Segers, H. & Martens, K.), no. 198 in Developments in Hydrobiology. Springer Netherlands. ISBN 978-1-4020-8258-0 978-1-4020-8259-7, pp. 545–567. URL http://link.springer.com/chapter/10.1007/978-1-4020-8259-7$_5$3.

8.

Rosenzweig, M. L. (2001). Loss of speciation rate will impoverish future diversity. *Proceedings of the National Academy of Sciences*, 98, 5404–5410.

9.

Rosindell, J. & Phillimore, A. B. (2011). A unified model of island biogeography sheds light on the zone of radiation. *Ecology Letters*, 14, 552–560.

10.

Seehausen, O., Alphen, J. J. M. v. & Witte, F. (1997). Cichlid Fish Diversity Threatened by Eutrophication That Curbs Sexual Selection. *Science*, 277, 1808–1811.

11.

Vadeboncoeur, Y., McIntyre, P. B. & Zanden, M. J. V. (2011). Borders of Biodiversity: Life at the Edge of the World's Large Lakes. *BioScience*, 61, 526–537.

12.

Vonlanthen, P., Bittner, D., Hudson, A. G., Young, K. A., Müller, R., Lundsgaard-Hansen, B., Roy, D., Di Piazza, S., Largiader, C. R. & Seehausen, O. (2012). Eutrophication causes speciation reversal in whitefish adaptive radiations. *Nature*, 482, 357–362.

13.

Vörösmarty, C. J., McIntyre, P. B., Gessner, M. O., Dudgeon, D., Prusevich, A., Green, P., Glidden, S., Bunn, S. E., Sullivan, C. A., Liermann, C. R. & Davies, P. M. (2010). Global threats to human water security and river biodiversity. *Nature*, 467, 555–561.

14.

Wagner, C. E., Harmon, L. J. & Seehausen, O. (2014). Cichlid species-area relationships are shaped by adaptive radiations that scale with area. *Ecology Letters*, 17, 583–592.

Symbol	Explanation
$\mathcal{N}$	Number of nodes
$\mathcal{S}$	Number of lakes
$\mathcal{E}_i$	Elevation lake i
$\mathrm{J_i}$	Number of individuals in lake i
$\mathrm{m_{ij}^{k}}$	Dispersal from lake j to lake i for species k
$\mathcal{K}$	Probability of cladogenesis initiation
$\mathcal{G}$	Number of generations retarding anagenesis
τ	Number of generations to have cladogenesis speciation
m	Intensity of dispersal
ν	Dispersal rate from the regional species pool
c	Upstream cost
λ	Local birth rate of metacommunity
$\mathrm{d_{ij}}$	Geographical distance between lake or node i and j

Table 1. Symbols used and parameter values

6 Tables

7 Figure Legends

Figure 1. 3D dendritic networks. FIGURE SHOWN ONE OF THE 3D EMPIRICAL DENDRITIC NETWORKS ANALYZED

Figure 2. Flow chart representing the ABC method to infer the mechanisms that best predict biodiversity patterns in 3D dendritic networks.

Figure 3. Upstream cost in 3D dendritic network. Probability to disperse (y-axis) from lake j to lake i as a function of distance (x-axis) given equal volume for lake i and j and with 500m and 10m elevation, respectively. Upstream cost increase from blue line (c = 0.0001) to red (c = 0.001), and to green (c = 0.01). At short distances, the larger the upstream migration cost, the lower the migration probability between these two lakes. At large distances the effect of the cost descreases and distances becomes more important.

8 Figures

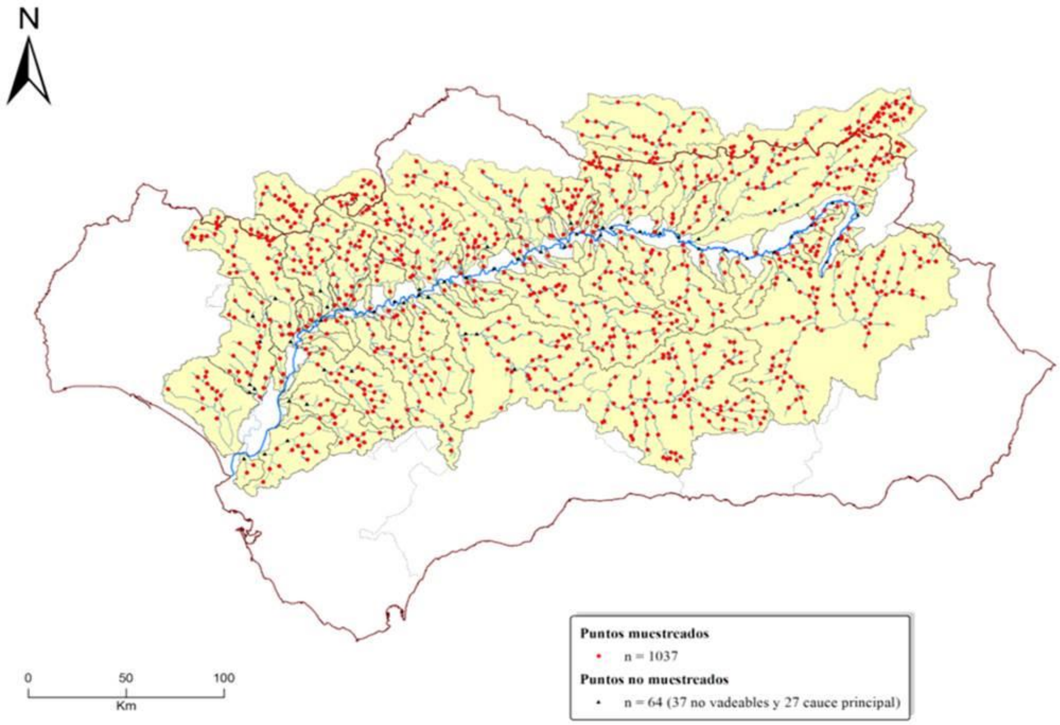

Figure 1

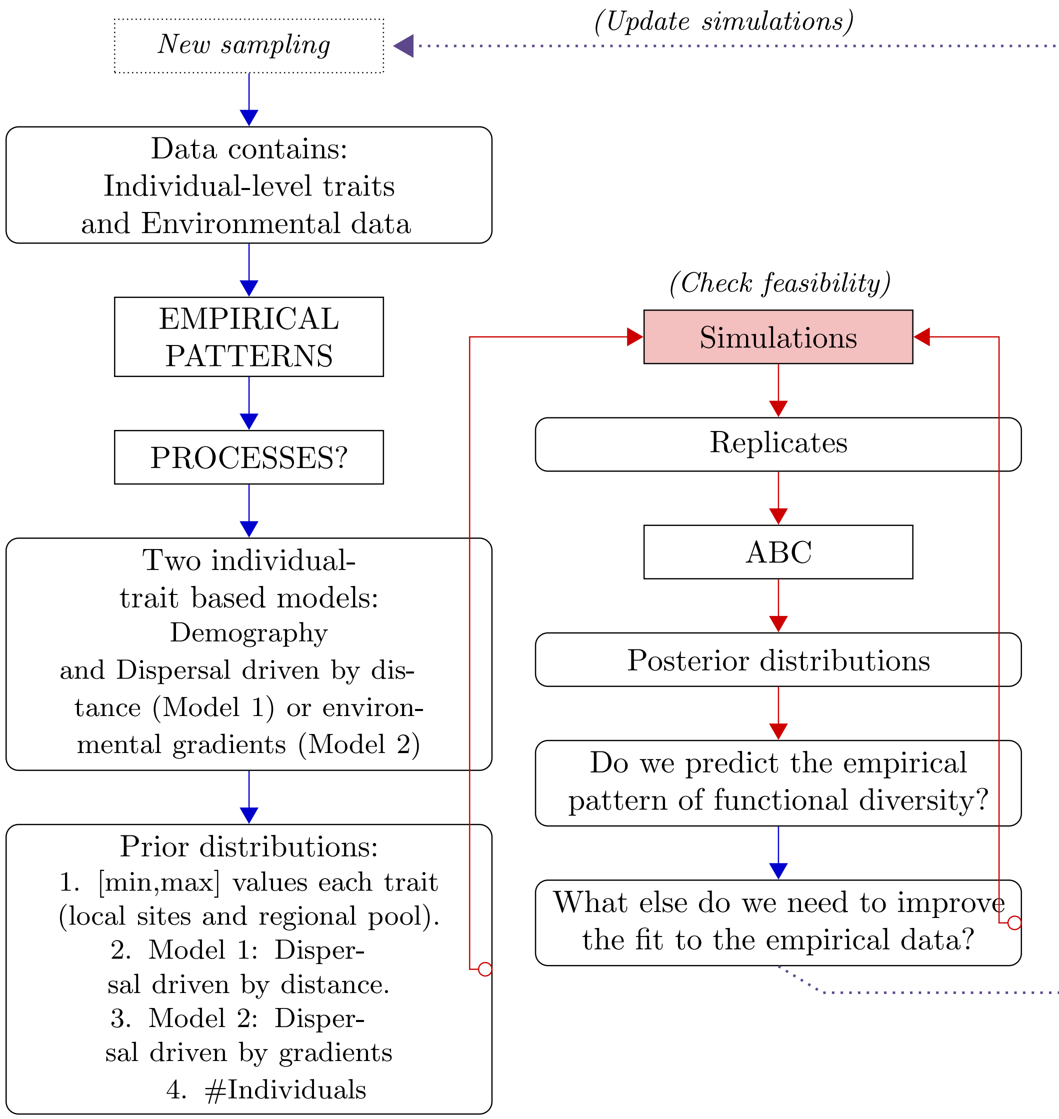
(Update simulations)
New sampling
Data contains:
Individual-level traits
and Environmental data
EMPIRICAL
PATTERNS
PROCESSES?
Two individual-
trait based models:
Demography
and Dispersal driven by dis-
tance (Model 1) or environ-
mental gradients (Model 2)
Prior distributions:
1. [min,max] values each trait
(local sites and regional pool).
2. Model 1: Disper-
sal driven by distance.
3. Model 2: Disper-
sal driven by gradients
4. #Individuals
(Check feasibility)
Simulations
Replicates
ABC
Posterior distributions
Do we predict the empirical
pattern of functional diversity?
What else do we need to improve
the fit to the empirical data?

Figure 2

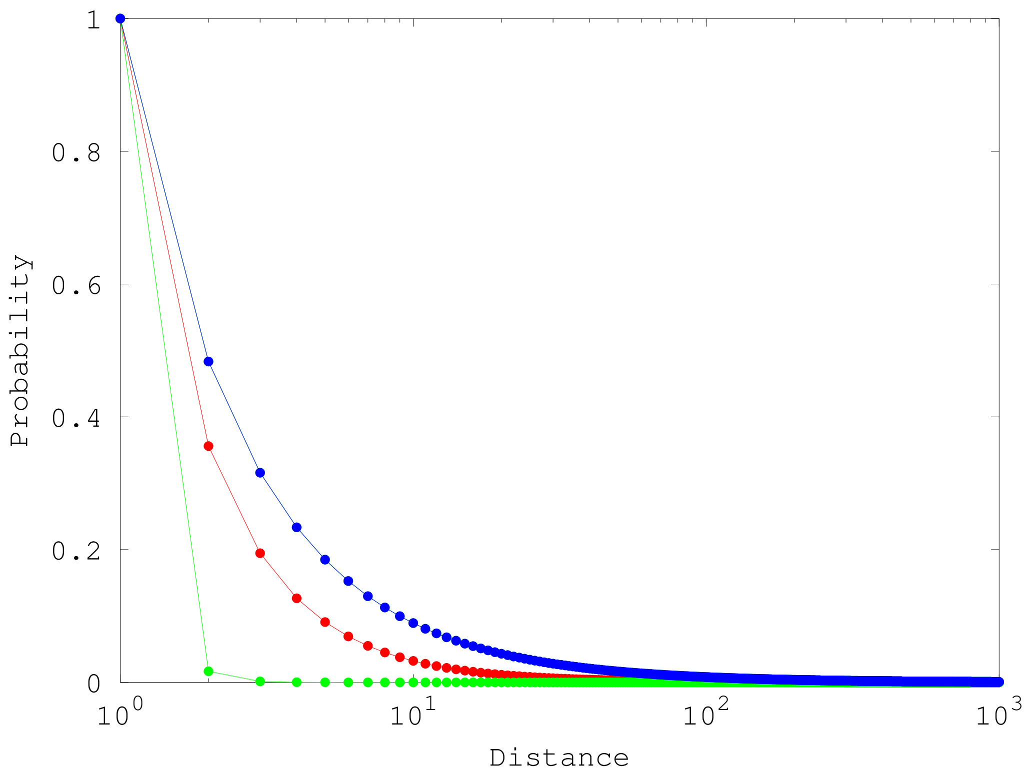

Figure 3